

# Linux Lab 3

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## Question: 1

```
[csf@localhost ~]$ sudo adduser Alice
[sudo] password for csf:
[csf@localhost ~]$ tail -1 /etc/passwd
Alice:x:1004:1005::/home/Alice:/bin/bash
[csf@localhost ~]$ sudo passwd Alice
Changing password for user Alice.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[csf@localhost ~]$ █
```

## Question: 2

```
[csf@localhost ~]$ touch course.txt
[csf@localhost ~]$ ls -l course.txt
-rw-r--r--. 1 csf csf 0 Aug 29 14:53 course.txt
```

## Question: 3

```
[csf@localhost ~]$ stat -c %a course.txt
644
```

#### Question: 4

```
[csf@localhost ~]$ sudo adduser Bob
[csf@localhost ~]$ sudo passwd Bob
Changing password for user Bob.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[csf@localhost ~]$ tail -1 /etc/passwd
Bob:x:1005:1006:./home/Bob:/bin/bash
[csf@localhost ~]$ █
```

```
[csf@localhost ~]$ ls -l course.txt
-rw-r--r--. 1 csf csf 0 Aug 29 14:53 course.txt
[csf@localhost ~]$ sudo chown Bob course.txt
[sudo] password for csf:
[csf@localhost ~]$ ls -l course.txt
-rw-r--r--. 1 Bob csf 0 Aug 29 14:53 course.txt
[csf@localhost ~]$ █
```

```
[csf@localhost ~]$ sudo chgrp Bob course.txt
[csf@localhost ~]$ ls -l course.txt
-rw-r--r--. 1 Bob Bob 0 Aug 29 14:53 course.txt
[csf@localhost ~]$
```

#### Question: 5

```
[csf@localhost ~]$ umask
0022
[csf@localhost ~]$ touch f1.txt
[csf@localhost ~]$ ls -l f1.txt
-rw-r--r--. 1 csf csf 0 Aug 29 15:10 f1.txt
[csf@localhost ~]$ umask 0027
[csf@localhost ~]$ touch f2.txt
[csf@localhost ~]$ ls -l f1.txt f2.txt
-rw-r--r--. 1 csf csf 0 Aug 29 15:10 f1.txt
-rw-r-----. 1 csf csf 0 Aug 29 15:10 f2.txt
[csf@localhost ~]$
```

**Question: 6**

```
[csf@localhost ~]$ touch hello_world.txt
[csf@localhost ~]$ ls -l hello_world.txt
-rw-r--r--. 1 csf csf 0 Aug 29 15:12 hello_world.txt
[csf@localhost ~]$ getfacl hello_world.txt
# file: hello_world.txt
# owner: csf
# group: csf
user::rw-
group::r--
other::r--

[csf@localhost ~]$
```

**Question: 7**

```
[root@localhost /]# cat > hello_world.txt
Hello World!
^C
[root@localhost /]# cat hello_world.txt
Hello World!
[root@localhost /]# ls -l hello_world.txt
-rw-r--r--. 1 root root 13 Aug 29 15:21 hello_world.txt

[root@localhost /]# setfacl -m u:Bob:r hello_world.txt
[root@localhost /]# ls -l hello_world.txt
-rw-r--r--+ 1 root root 13 Aug 29 15:21 hello_world.txt
[root@localhost /]# su Bob
[Bob@localhost /]$ cat hello_world.txt
Hello World!
[Bob@localhost /]$ █
```

#### Question: 8

```
[root@localhost /]# ls -l hello_world.txt
-rw-r--r--+ 1 root root 13 Aug 29 15:21 hello_world.txt
[root@localhost /]# getfacl hello_world.txt
# file: hello_world.txt
# owner: root
# group: root
user::rw-
user:Bob:r--
group::r--
mask::r--
other::r--
```

```
[root@localhost /]# setfacl -m u:Bob:--- hello_world.txt
[root@localhost /]# getfacl hello_world.txt
# file: hello_world.txt
# owner: root
# group: root
user::rw-
user:Bob:---
group::r--
mask::r--
other::r--

[root@localhost /]# █
```

#### Question: 9

```
[csf@localhost ~]$ mkdir directory1
[csf@localhost ~]$ ls -l
total 16
-rw-r--r--. 1 Bob Bob 0 Aug 29 14:53 course.txt
drwxr-xr-x. 2 csf csf 4096 Aug 29 15:29 directory1
-rw-r--r--. 1 csf csf 0 Aug 29 15:10 f1.txt
-rw-r-----. 1 csf csf 0 Aug 29 15:10 f2.txt
-rw-r--r--. 1 csf csf 6503 Aug 29 12:48 file.txt
-rw-r--r--+ 1 csf csf 0 Aug 29 15:12 hello_world.txt
```

#### Question: 10

```
[csf@localhost ~]$ ll
total 4
drwxr-xr-x. 2 Alice Alice 4096 Aug 29 15:29 directory1
[csf@localhost ~]$ sudo chown Bob directory1/
[csf@localhost ~]$ ll
total 4
drwxr-xr-x. 2 Bob Alice 4096 Aug 29 15:29 directory1
[csf@localhost ~]$ sudo chgrp Bob directory1/
[csf@localhost ~]$ ll
total 4
drwxr-xr-x. 2 Bob Bob 4096 Aug 29 15:29 directory1
[csf@localhost ~]$ █
```

Question: 11

```
[root@localhost /]# ls -l confidential.txt
-rw-rw----. 1 Alice Alice 0 Aug 29 15:45 confidential.txt
[root@localhost /]# su Bob
[Bob@localhost /]$ cat confidential.txt
cat: confidential.txt: Permission denied
[Bob@localhost /]$ su
Password:
[root@localhost /]# getfacl confidential.txt
# file: confidential.txt
# owner: Alice
# group: Alice
user::rw-
group::rw-
other::---

[root@localhost /]# setfacl -m u:Bob:r confidential.txt
[root@localhost /]# getfacl confidential.txt
# file: confidential.txt
# owner: Alice
# group: Alice
user::rw-
user:Bob:r--
group::rw-
mask::rw-
other::---

[root@localhost /]# su Bob
[Bob@localhost /]$ cat confidential.txt
[Bob@localhost /]$ ls -l confidential.txt
-rw-rw----+ 1 Alice Alice 0 Aug 29 15:45 confidential.txt
[Bob@localhost /]$ █
```

**Question: 12**

```
[root@localhost /]# getfacl confidential.txt
# file: confidential.txt
# owner: Alice
# group: Alice
user::rw-
user:Bob:r--
group::rw-
mask::rw-
other::---

[root@localhost /]# setfacl -x u:Bob confidential.txt
[root@localhost /]# getfacl confidential.txt
# file: confidential.txt
# owner: Alice
# group: Alice
user::rw-
group::rw-
mask::rw-
other::---

[root@localhost /]# █
```

**Question: 13**

```
[root@localhost /]# ll report.txt
-rw-----. 1 Alice Alice 15 Aug 29 15:55 report.txt
[root@localhost /]# getfacl report.txt
# file: report.txt
# owner: Alice
# group: Alice
user::rw-
group::---
other::---

[root@localhost /]# setfacl -m u:Bob:rw-,u:charlie:r-- report.txt
[root@localhost /]# getfacl report.txt
# file: report.txt
# owner: Alice
# group: Alice
user::rw-
user:Bob:rw-
user:charlie:r--
group::---
mask::rw-
other::---

[root@localhost /]# █
```

#### Question 14

```
[root@localhost /]# ls -ld shared/
drwx-----. 2 Alice Alice 4096 Aug 29 16:03 shared/
[root@localhost /]# setfacl -m u:Bob:r-x shared
[root@localhost /]# ls -ld shared/
drwxr-x---+ 2 Alice Alice 4096 Aug 29 16:03 shared/
[root@localhost /]# getfacl shared/
# file: shared/
# owner: Alice
# group: Alice
user::rwx
user:Bob:r-x
group:---
mask::r-x
other:---

[root@localhost /]# █
```